Sheaf-Theoretic Methods in Groundwater Hydrogeochemistry: A Mathematical Framework for Inverse Geochemical Modeling

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Abstract

We present a comprehensive mathematical framework for solving inverse problems in groundwater hydrogeochemistry using sheaf-theoretic principles. The method determines optimal transport processes (evaporation, mixing) and geochemical reactions that explain observed chemical evolution along flow paths in aquifer networks. The framework combines weighted least squares optimization, sparse LASSO regression with coordinate descent, thermodynamic constraints from PHREEQC, and isotope hydrogeology to produce parsimonious, physically consistent models. We establish theoretical results on the optimality of transport models, convergence properties of the coordinate descent algorithm, and probabilistic methods for inferring flow network topology from incomplete hydraulic head data. The framework integrates linear algebra, convex optimization, graph theory, and geochemical thermodynamics into a unified computational approach suitable for practical groundwater management applications. Similar to other open-source projects, the implementation is available at `https://github.com/dabdul-wahab1988/Hydrosheaf`.

1 Introduction

Groundwater hydrogeochemistry seeks to understand the chemical evolution of water as it flows through aquifer systems. The inverse problem—determining which combination of transport processes (mixing, evaporation) and geochemical reactions (mineral dissolution/precipitation, redox reactions, ion exchange) explains observed chemical changes—is fundamental to aquifer characterization, contamination assessment, and water resource management.

1.1 Problem Motivation and Formulation

Consider a groundwater flow network represented as a directed graph $G = (V, E)$, where vertices V represent sampling locations (wells) and edges E represent inferred flow connections. At each vertex $v \in V$, we observe a vector of ion concentrations $\mathbf{x}_v \in \mathbb{R}^n$ (default $n = 10$ columns: Ca, Mg, Na, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4 in mmol/L).

1.1.1 Input-Output Contract

The framework takes as input:

- A samples CSV containing observed ion concentrations $\{\mathbf{x}_v\}_{v\in V}$ (mmol/L) at each node, with nodes identified by `site_id` and `sample_id`. The default ion vector uses $n = 10$ columns (`Ca, Mg, Na, HCO3, Cl, SO4, NO3, F, Fe, PO4`) as defined in `hydrosheaf/config.py` and validated by the CLI schema in `hydrosheaf/cli.py`. The order can be overridden with `--ion-order` (must provide exactly 10 comma-separated ion names).

- Supporting bulk water fields used for diagnostics and penalties: `EC`, `TDS`, and `pH` (*required by the current CLI schema in* `hydrosheaf/cli.py`).

- Optional isotope data for process discrimination, including water isotopes (δ^{18}O, δ^2H; default columns `18O`, `2H`, configurable via `--isotope-d18o-key`, `--isotope-d2h-key`) and nitrate isotopes (δ^{15}N–NO_3, δ^{18}O–NO_3; default columns `d15N`, `d18O_NO3` in `hydrosheaf/config.py`).

- Hydraulic head measurements or topographic elevations for network inference (e.g., `head_meas`, `dtw`, `elevation`; configured by `--edge-*-key` flags). Optional coordinates (`x,y` or `lon,lat`) and depth/screen metadata support 3D inference.

- A dictionary of candidate mineral reactions with stoichiometric coefficients

- Endmember compositions (e.g., rainfall, river water) for mixing hypotheses

- Optional temporal time-series CSV when `--temporal-enabled` is used. The temporal contract is a single long-form CSV (`hydrosheaf/temporal/time_series.py`) with a node id column (`site_id` preferred; `node_id` and `sample_id` are attempted as fallbacks), a `timestamp` column (default parseable formats include `YYYY-MM-DD`), and any subset of ion/tracer columns used for temporal fitting (e.g., `Cl`, `18O`, `2H`).

- Optional physics priors when mesh-based flow/particle tracking outputs are available:
  - (i) a CSV/JSON file of (u, v) edges with optional `p_uv` and travel-time summaries (recommended columns: `tt_mean_days`, `tt_std_days`, `tt_p10_days`, `tt_p90_days`; see `hydrosheaf/physics/priors.py`)
  - (ii) a MODPATH endpoints file ingested via FloPy (see `hydrosheaf/physics/modpath.py`)

  In both cases, u and v must match node identifiers in the samples.

The framework produces as output:

- For each edge $(u, v) \in E$: the optimal transport model (evaporation factor γ or mixing fraction f)

- A sparse set of active reactions with quantified extents $\mathbf{z}$ (positive = dissolution, negative = precipitation)

- Model fit quality metrics (R^2, residual norms, Boltzmann probabilities)

- Diagnostic flags (charge balance errors, thermodynamic violations, isotope consistency)

1.1.2 Problem Difficulty

For each directed edge $(u, v) \in E$, we seek to determine:

1. The dominant transport process (evaporation or mixing with an endmember)
2. The extent of mineral reactions that explain residual chemical changes
3. The consistency of this explanation with thermodynamic, isotopic, and mass balance constraints

This inverse problem is fundamentally ill-posed due to several factors:

- **Non-uniqueness**: Multiple combinations of transport and reactions can produce similar chemical evolution. For example, an increase in Ca and SO_4 could be explained by gypsum dissolution, evaporation, or mixing with saline water—or some combination thereof.
- **Underdetermination**: With typically 8–10 major-ion constraints per sample (default $n = 10$ in the implementation) and 20–30 candidate reactions, the system is underdetermined ($m \gg n$). Infinitely many reaction combinations satisfy the mass balance equations.
- **Measurement uncertainty**: Ion concentrations have analytical uncertainties (typically 2-5%), and hydraulic head measurements are noisy, leading to uncertain flow network topology.
- **Scale mismatch**: Laboratory-derived thermodynamic and kinetic parameters may not accurately represent field-scale heterogeneous aquifer conditions.

1.1.3 Limitations of Existing Methods

Traditional geochemical modeling tools approach this problem differently:

- **Inverse geochemical mass-balance (PHREEQC, NETPATH)** can generate multiple admissible inverse models (sets of phase/reaction transfers) for the same pair of waters, and users typically add additional constraints to reduce the solution set [6, 7]. When many candidate phases are allowed, the number of feasible combinations can grow combinatorially (e.g., $2^{30} \approx 10^9$ subsets for 30 independent candidates), making manual screening and expert judgment a practical bottleneck.
- **Forward modeling emphasis (GWB)**: many workflows focus on forward reaction-path or speciation modeling, where reactions/processes are specified a priori and simulations are compared to observations, rather than solving an automated inverse fitting problem [8].
- **Thermodynamic guidance and non-uniqueness**: classic mass-balance tools such as NETPATH are primarily designed for net-reaction stoichiometry along a user-defined flow path, with more limited thermodynamic feasibility screening than equilibrium-based inverse approaches [7]. PHREEQC extends inverse modeling with uncertainty handling and stronger equilibrium constraints [6], but inverse solutions can still be non-unique and neither tool provides built-in sparsity/parsimony-based model selection (e.g., L1-regularized selection) to automatically prefer the simplest adequate explanation.

- **Flow-path/network specification**: the analyst must define the flow path or network topology externally (choice of upstream/downstream connections, candidate endmembers, and sampling pairing) [7]. This is a major limitation in regional settings with sparse head data and uncertain connectivity.

None of these tools provide automated model selection with explicit parsimony principles, probabilistic network inference, or integrated nitrate source discrimination using $\delta^{15}N{-}NO_3$ and $\delta^{18}O{-}NO_3$ constraints. While regularization and machine-learning-assisted inverse modeling are active research directions, they are not part of classic packaged toolchains and remain largely bespoke workflows in the literature [9, 10].

1.1.4 Solution Strategy

To obtain meaningful, physically plausible solutions, we employ three complementary strategies:

- **Sparsity regularization**: Prefer explanations involving fewer reactions (Occam's razor). The L1 penalty in LASSO regression automatically selects a sparse subset of reactions from the candidate dictionary.
- **Thermodynamic constraints**: Enforce mineral equilibrium bounds via saturation indices computed by PHREEQC. This eliminates geochemically impossible solutions (e.g., precipitation in undersaturated conditions).
- **Multi-physics integration**: Incorporate isotope data ($\delta^{18}O$, δ^2H) and electrical conductivity to discriminate between processes that produce similar ion signatures. Evaporation and mixing, for example, have distinct isotopic fingerprints.

These strategies transform the ill-posed inverse problem into a well-regularized optimization problem with unique, interpretable solutions.

1.2 Sheaf-Theoretic Perspective

The sheaf-theoretic viewpoint interprets the groundwater network as a cellular complex where local data (ion concentrations at wells) must satisfy global consistency conditions (mass balance, charge balance, thermodynamic equilibrium). While we do not develop full sheaf cohomology here, the computational framework enforces consistency through residual minimization along edges, analogous to minimizing the discrepancy in sheaf-theoretic data structures.

1.3 Document Roadmap

The remainder of this document is organized as follows: Section 2 presents the core mathematical framework, including the optimization problem formulation and two-stage solution strategy. Section 3 develops probabilistic methods for inferring flow network topology from hydraulic head data. Section 4 establishes theoretical results on optimal transport models (evaporation and mixing). Section 5 presents the sparse reaction fitting algorithm using LASSO regression and coordinate descent. Section 6 describes thermodynamic constraints derived from saturation indices. Section 7 introduces isotope-based penalties for process discrimination. Section 8 presents the specialized module for nitrate source discrimination using compositional data analysis and Bayesian evidence. Section 9 summarizes the complete algorithm and computational complexity. Section 10 presents validation methodology and representative applications. Section 11 compares our approach to existing commercial software. Section 12 discusses advanced extensions including reactive transport, 3D networks, temporal dynamics, and uncertainty quantification. Section 13 concludes with a summary and future perspectives.

1.4 Notation

The following table summarizes the principal mathematical symbols used throughout this document.

Table 1: Principal Mathematical Notation

Symbol	Description
$G = (V, E)$	Directed graph representing flow network (vertices = wells, edges = flow paths)
$\mathbf{x}_v \in \mathbb{R}^n$	Ion concentration vector at node v (implementation default $n =$ 10: Ca, Mg, Na, HCO_3, Cl, SO_4, NO_3, F, Fe, PO_4)
$\boldsymbol{\theta}$	Transport model parameters (γ for evaporation, f for mixing)
$\gamma \geq 1$	Evaporation concentration factor
$f \in [0, 1]$	Mixing fraction with endmember $\mathbf{x}_{\text{end}}$
$S \in \mathbb{R}^{n \times m}$	Stoichiometric matrix (m reactions, n ions)
$\mathbf{z} \in \mathbb{R}^m$	Reaction extent vector (positive = dissolution, negative = precipitation)
$\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$	Diagonal weight matrix (typically inverse variance)
$\lambda > 0$	L1 regularization parameter (controls sparsity)
$\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^m$	Lower and upper bounds on reaction extents (from saturation indices)
$A(\boldsymbol{\theta}), \mathbf{b}(\boldsymbol{\theta})$	Affine transformation representing transport process
$\mathbf{r} \in \mathbb{R}^n$	Residual vector after transport ($\mathbf{x}_v - A\mathbf{x}_u - \mathbf{b}$)
$\mathcal{P}_*$	Penalty functions (isotope, EC/TDS, charge balance, thermodynamic)
SI	Saturation index $= \log_{10}(\text{IAP}/K_{sp})$
h_i, h_j	Hydraulic heads at nodes i, j
$p(i \to j)$	Probability of flow from node i to j
$\delta^{18}\text{O}, \delta^2\text{H}$	Stable water isotope ratios (permil relative to VSMOW)
$d = \delta^2\text{H} - 8 \cdot \delta^{18}\text{O}$	Deuterium excess (permil)
$\mathcal{S}(\alpha, \tau)$	Soft-thresholding operator for LASSO
T	Number of coordinate descent iterations to convergence
$\lvert\mathcal{C}\rvert$	Number of candidate transport models
$\mathcal{A} = \{j : z_j^* \neq 0\}$	Active set of reactions (non-zero extents)
R^2	Coefficient of determination (fit quality metric)

Throughout the document, vectors are denoted in boldface (e.g., $\mathbf{x}, \mathbf{z}$), scalars in italic (e.g., γ, λ), and matrices in boldface uppercase (e.g., $\mathbf{W}, S$). All concentration units are mmol/L unless otherwise specified.

2 Mathematical Framework

2.1 Core Optimization Problem

For each edge $(u, v) \in E$, we solve the following constrained optimization problem:

$$\begin{aligned} \min_{\boldsymbol{\theta},\mathbf{z}} \quad & \mathcal{L}(\boldsymbol{\theta},\mathbf{z}) = \frac{1}{2}\|\mathbf{x}_v - \mathbf{x}_{\text{pred}}\|_{\mathbf{W}}^2 + \lambda\|\mathbf{z}\|_1 \\ & \qquad + \mathcal{P}_{\text{EC/TDS}}(\boldsymbol{\theta},\mathbf{z}) + \mathcal{P}_{\text{iso}}(\boldsymbol{\theta}) + \mathcal{P}_{\text{Gibbs}}(\boldsymbol{\theta}) + \mathcal{P}_{\text{cons}}(\mathbf{z}) \\ \text{subject to} \quad & \boldsymbol{\ell} \le \mathbf{z} \le \mathbf{u}, \quad \boldsymbol{\theta} \in \Theta \end{aligned} \tag{1}$$

where:

- $\mathbf{x}_v \in \mathbb{R}^n$: observed ion concentration vector at downstream node v
- $\mathbf{x}_{\text{pred}} = A(\boldsymbol{\theta})\mathbf{x}_u + \mathbf{b}(\boldsymbol{\theta}) + S\mathbf{z}$: predicted concentration
- $\boldsymbol{\theta}$: transport model parameters (evaporation factor or mixing fraction)
- $A(\boldsymbol{\theta}), \mathbf{b}(\boldsymbol{\theta})$: affine transformation representing transport
- $S \in \mathbb{R}^{n\times m}$: stoichiometric matrix (m reactions)
- $\mathbf{z} \in \mathbb{R}^m$: reaction extent vector (positive = dissolution, negative = precipitation)
- $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$: weight matrix (typically inverse variance or ion importance)
- $\lambda > 0$: L1 regularization parameter promoting sparsity
- $\mathcal{P}_*$: penalty functions enforcing physical constraints
- $\boldsymbol{\ell}, \mathbf{u} \in \mathbb{R}^m$: thermodynamically-derived bounds on reaction extents

The penalty terms are defined as follows:

$$\mathcal{P}_{\text{EC/TDS}}(\mathbf{z}) = \eta_1\|\text{EC}_{\text{pred}}(\mathbf{z}) - \text{EC}_{\text{obs}}\|^2 + \eta_2\|\text{TDS}_{\text{pred}}(\mathbf{z}) - \text{TDS}_{\text{obs}}\|^2 \tag{2}$$

$$\mathcal{P}_{\text{Gibbs}}(\boldsymbol{\theta}) = \begin{cases} 0 & \text{if } \boldsymbol{\theta} \in \Theta_{\text{phys}} \\ \infty & \text{otherwise} \end{cases} \tag{3}$$

$$\mathcal{P}_{\text{cons}}(\mathbf{z}) = \eta_3(\text{ChargeBalance}(\mathbf{x}_{\text{pred}}))^2 \tag{4}$$

where Θ_{phys} represents the physically valid parameter space (e.g., $\gamma \ge 1, f \in [0, 1]$).

Definition 2.1 (Weighted Norm)**.** For a positive definite diagonal matrix $\mathbf{W} = \text{diag}(w_1, \ldots, w_n)$ with $w_i > 0$, the weighted Euclidean norm is:

$$\|\mathbf{r}\|_{\mathbf{W}}^2 = \mathbf{r}^\top\mathbf{W}\mathbf{r} = \sum_{i=1}^{n} w_i r_i^2 \tag{5}$$

This norm emphasizes ions with larger weights, typically chosen as $w_i = 1/\sigma_i^2$ for measurement uncertainty σ_i, or based on geochemical significance.

2.2 Two-Stage Optimization Strategy

Problem (1) is solved in two stages:

1. **Transport Optimization**: Fix $\mathbf{z} = \mathbf{0}$ and optimize over $\boldsymbol{\theta}$ to find the best transport model:

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\Theta} \|\mathbf{x}_v - A(\boldsymbol{\theta})\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta})\|_{\mathbf{W}}^2 + \mathcal{P}_{\text{iso}}(\boldsymbol{\theta}) + \mathcal{P}_{\text{Gibbs}}(\boldsymbol{\theta}) \tag{6}$$

2. **Reaction Optimization**: Fix $\boldsymbol{\theta} = \boldsymbol{\theta}^*$, compute residual $\mathbf{r} = \mathbf{x}_v - A(\boldsymbol{\theta}^*)\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta}^*)$, and solve:

$$\mathbf{z}^* = \arg\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{r} - S\mathbf{z}\|_{\mathbf{W}}^2 + \lambda\|\mathbf{z}\|_1 + \mathcal{P}_{\text{EC/TDS}}(\mathbf{z}) + \mathcal{P}_{\text{cons}}(\mathbf{z}) \quad \text{s.t.} \quad \boldsymbol{\ell} \leq \mathbf{z} \leq \mathbf{u} \tag{7}$$

This decomposition exploits the structure of the problem: transport models have few parameters (1-2) amenable to exhaustive search or analytic solution, while reaction fitting is high-dimensional but sparse.

Remark 2.2 (Computational Efficiency). The two-stage strategy reduces computational cost dramatically. If we jointly optimized over $(\boldsymbol{\theta}, \mathbf{z})$, we would face a non-convex mixed continuous-discrete problem with $O(|\mathcal{C}| \cdot 2^m)$ potential model combinations (since each reaction can be active or inactive). By decoupling, we reduce this to $O(|\mathcal{C}|)$ transport evaluations plus one convex optimization in $\mathbb{R}^m$.

3 Graph-Theoretic Edge Inference

Before applying the transport and reaction models described in subsequent sections, we must first construct the flow network $G = (V, E)$ from observational data. In groundwater systems, flow directions are inferred from hydraulic head measurements, which are subject to measurement uncertainty and spatial interpolation errors. This section develops probabilistic methods for edge inference that appropriately propagate uncertainty into the network topology.

3.1 Probabilistic Edge Weights

In groundwater networks, flow directions are inferred from hydraulic head measurements. Due to measurement uncertainty and spatial interpolation, we assign probabilistic weights to potential edges.

Given hydraulic heads h_i, h_j with uncertainties σ_i, σ_j at nodes i, j separated by distance d_{ij}, the head difference is:

$$\Delta h_{ij} = h_i - h_j \sim \mathcal{N}(\mu_{ij}, \sigma_{ij}^2) \tag{8}$$

where $\mu_{ij} = \hat{h}_i - \hat{h}_j$ and $\sigma_{ij} = \sqrt{\sigma_i^2 + \sigma_j^2}$ (assuming independence).

The probability of flow from i to j is:

$$p(i \to j) = \Phi(\mu_{ij}/\sigma_{ij}) \quad (\textit{Verified in}\texttt{tests/test\_graph\_probabilistic.py}) \tag{9}$$

where Φ is the standard normal cumulative distribution function.

3.2 Hydraulic Gradient Guard

To avoid spurious connections over large distances with small head differences (unrealistically flat gradients), we apply a gradient threshold. The hydraulic gradient is:

$$g_{ij} = \frac{|\Delta h_{ij}|}{d_{ij}} \tag{10}$$

If $g_{ij} < g_{\min}$ (e.g., $g_{\min} = 0.001$ or 1 m per 1000 m), we attenuate the edge probability:

$$p(i \to j) \leftarrow 0.5 + (p(i \to j) - 0.5) \cdot \frac{g_{ij}}{g_{\min}} \tag{11}$$

This pulls probabilities toward 0.5 (maximum uncertainty) when gradients are implausibly small.

3.3 Hierarchical Head Estimation

When direct head measurements are unavailable, we estimate head using secondary data and propagate uncertainty into edge direction probabilities. Hydrosheaf implements both a heuristic tiering scheme and a linear-Gaussian Bayesian model (`hydrosheaf/graph/head_inference.`

3.3.1 Tiered Data Model

1. **Tier A (Direct)**: Measured head h with uncertainty $\sigma = 0.5$ m
2. **Tier B (Depth to Water)**: $h = h_{\text{surface}} - d_{\text{water}}$ with $\sigma = \sqrt{\sigma^2_{\text{surface}} + \sigma^2_{\text{water}}}$
3. **Tier C (Topography)**: $h \approx h_{\text{surface}}$ with large uncertainty $\sigma = 10$ m

This hierarchical approach maximizes network coverage while appropriately propagating uncertainty.

3.3.2 Bayesian Linear-Gaussian Posterior

Let h_i denote the latent hydraulic head at node i and let μ_{dtw} be a shared mean depth-to-water parameter. With elevation z_i available, we use a soft topographic prior:

$$h_i + \mu_{dtw} \approx z_i \quad \text{with} \quad \sigma_{\text{topo}} \tag{12}$$

and incorporate observations of head (Tier A) and elevation-minus-depth-to-water (Tier B) as independent Gaussian measurements. The resulting posterior is multivariate normal and can be computed in closed form by assembling a precision matrix and solving linear systems (no MCMC required). When `--edge-head-inference bayesian_mcmc` is selected and PyMC is available, a sampling-based posterior may be used; otherwise the implementation falls back to the exact linear solve.

3.4 MAP-Weighted Edge Selection

When multiple candidate downstream neighbors are plausible for a node, Hydrosheaf supports MAP-style selection that combines chemistry fit quality with probabilistic edge priors. Given a candidate edge $(u \to v)$ with inferred probability p_{uv} and geochemical objective J_{uv} (the same `objective_score` used for transport+reaction fitting), we define:

$$J_{uv}^{MAP} = J_{uv} + w \cdot \left(-\log(p_{uv} + \varepsilon)\right), \tag{13}$$

and keep the top-k edges per upstream node minimizing J_{uv}^{MAP}. This is selected with `--infer-edges-methodprobabilistic_map` and parameters `--edge-map-weight`, `--edge-map-candi`
`--edge-map-p-min` (see `hydrosheaf/inference/network_fit.py`).

3.5 Optional Physics Priors for Edges and Travel Time

When mesh-based flow/particle tracking outputs exist (e.g., MODFLOW/MODPATH), they can be imported as priors on $(u \to v)$ edges:

- **Connectivity**: set p_{uv} from particle connectivity fractions
- **Travel time**: set τ_{uv} (mean and quantiles) to override or inform residence time estimation

Hydrosheaf supports a file-based contract (`--physics-priors`) and a FloPy-based MODPATH endpoints reader (`--modpath-endpoints`) implemented in `hydrosheaf/physics/priors.py` and `hydrosheaf/physics/modpath.py`. MODFLOW/MODPATH and FloPy are standard tools in groundwater modeling workflows [11, 12, 13].

With the network topology established, we now proceed to model the geochemical processes occurring along each edge.

4 Transport Models

4.1 Evaporation Model

The evaporation model assumes conservative scaling of all solute concentrations:

$$\mathbf{x}' = \gamma \mathbf{x}_u, \quad \gamma \geq 1 \tag{14}$$

where γ is the concentration factor. Physically, γ represents the fraction of water lost to evapotranspiration: if $\gamma = 2$, half the water volume was lost.

Theorem 4.1 (Optimal Evaporation Factor)**.** *For the weighted least squares problem:*

$$\min_{\gamma \geq 1} \|\mathbf{x}_v - \gamma \mathbf{x}_u\|_{\mathbf{W}}^2 \tag{15}$$

the optimal solution is:

$$\gamma^* = \max\left\{1, \frac{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v}{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u}\right\} \tag{16}$$

Proof. Expanding the objective function:

$$f(\gamma) = \|\mathbf{x}_v - \gamma \mathbf{x}_u\|_{\mathbf{W}}^2 = (\mathbf{x}_v - \gamma \mathbf{x}_u)^\top \mathbf{W} (\mathbf{x}_v - \gamma \mathbf{x}_u) \tag{17}$$

$$= \mathbf{x}_v^\top \mathbf{W} \mathbf{x}_v - 2\gamma \mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v + \gamma^2 \mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u \tag{18}$$

This is a quadratic function in γ. Taking the derivative:

$$\frac{df}{d\gamma} = -2\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v + 2\gamma \mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u \tag{19}$$

Setting to zero yields:

$$\gamma_{\text{unconstrained}} = \frac{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v}{\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u} \tag{20}$$

Since $\mathbf{W}$ is positive definite, $\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u > 0$. The second derivative:

$$\frac{d^2 f}{d\gamma^2} = 2\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u > 0 \tag{21}$$

confirms this is a minimum. Enforcing the constraint $\gamma \geq 1$ (evaporation cannot dilute):

$$\gamma^* = \max\{1, \gamma_{\text{unconstrained}}\} \tag{22}$$

□

4.2 Single-Endmember Mixing Model

The mixing model represents dilution or mixing with a distinct water type (e.g., rainfall, river water):

$$\mathbf{x}' = (1 - f)\mathbf{x}_u + f \mathbf{x}_{\text{end}}, \quad f \in [0, 1] \tag{23}$$

where $\mathbf{x}_{\text{end}} \in \mathbb{R}^n$ is the endmember composition and f is the mixing fraction.

Theorem 4.2 (Optimal Mixing Fraction)**.** *For the weighted least squares problem:*

$$\min_{f\in[0,1]} \|\mathbf{x}_v - (1-f)\mathbf{x}_u - f\mathbf{x}_{end}\|^2_{\mathbf{W}} \tag{24}$$

the optimal solution is:

$$f^* = \max\left\{0, \min\left\{1, \frac{\mathbf{d}^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u)}{\mathbf{d}^\top \mathbf{W}\mathbf{d}}\right\}\right\} \tag{25}$$

where $\mathbf{d} = \mathbf{x}_{end} - \mathbf{x}_u$*.*

Proof. Let $\mathbf{d} = \mathbf{x}_{\text{end}} - \mathbf{x}_u$. Then:

$$\mathbf{x}' = \mathbf{x}_u + f\mathbf{d} \tag{26}$$

The objective becomes:

$$\begin{aligned} g(f) &= \|\mathbf{x}_v - \mathbf{x}_u - f\mathbf{d}\|^2_{\mathbf{W}} &(27)\\ &= (\mathbf{x}_v - \mathbf{x}_u)^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u) - 2f\mathbf{d}^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u) + f^2\mathbf{d}^\top \mathbf{W}\mathbf{d} &(28)\end{aligned}$$

Taking the derivative with respect to f:

$$\frac{dg}{df} = -2\mathbf{d}^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u) + 2f\mathbf{d}^\top \mathbf{W}\mathbf{d} \tag{29}$$

Setting to zero:

$$f_{\text{unconstrained}} = \frac{\mathbf{d}^\top \mathbf{W}(\mathbf{x}_v - \mathbf{x}_u)}{\mathbf{d}^\top \mathbf{W}\mathbf{d}} \tag{30}$$

The second derivative $\frac{d^2g}{df^2} = 2\mathbf{d}^\top \mathbf{W}\mathbf{d} > 0$ confirms this is a minimum. Projecting onto $[0, 1]$:

$$f^* = \begin{cases} 0 & \text{if } f_{\text{unconstrained}} < 0 \\ f_{\text{unconstrained}} & \text{if } f_{\text{unconstrained}} \in [0,1] \\ 1 & \text{if } f_{\text{unconstrained}} > 1 \end{cases} \tag{31}$$

which is equivalent to (25). □

4.3 Transport Model Selection

In practice, we evaluate multiple transport hypotheses (evaporation, mixing with various endmembers, or no transport) and select the best-fitting model. Let $\mathcal{C}$ be the set of candidate models indexed by c, each with optimal parameter $\boldsymbol{\theta}^*_c$ and objective value J_c. Model selection uses Boltzmann-weighted probabilities:

$$w_c = \exp(-(J_c - J_{\min})), \quad p_c = \frac{w_c}{\sum_{c'\in\mathcal{C}} w_{c'}} \tag{32}$$

where $J_{\min} = \min_{c\in\mathcal{C}} J_c$. This provides a soft model selection that accounts for model uncertainty.

Example 4.3 (Evaporation vs. Mixing Discrimination)**.** *This example is computationally verified in* `tests/test_doc_examples.py` (`test_example_4_4_transport_selection`).

Consider two nodes with concentrations (in mmol/L):

$$\mathbf{x}_u = [2.0, 1.0, 3.0, 5.0, 1.5, 2.0, 0.5, 0.1]^\top \quad \text{(Ca, Mg, Na, HCO}_3\text{, Cl, SO}_4\text{, NO}_3\text{, F)}$$
$$\mathbf{x}_v = [4.1, 2.0, 6.2, 10.1, 3.1, 4.0, 1.0, 0.2]^\top$$

and isotope data $(\delta^{18}\mathrm{O}_u, \delta^2\mathrm{H}_u) = (-5.0, -30.0)$ permil, $(\delta^{18}\mathrm{O}_v, \delta^2\mathrm{H}_v) = (-2.5, -22.0)$ permil.

Evaporation hypothesis: Using Theorem 4.1 with $\mathbf{W} = I$:

$$\gamma^* = \frac{\mathbf{x}_u^\top \mathbf{x}_v}{\mathbf{x}_u^\top \mathbf{x}_u} = \frac{2.0(4.1) + \cdots + 0.1(0.2)}{2.0^2 + \cdots + 0.1^2} = \frac{92.47}{45.51} \approx 2.03 \tag{33}$$

This predicts $\mathbf{x}_{\text{pred}} = 2.03\mathbf{x}_u$, with squared error $\|\mathbf{x}_v - \mathbf{x}_{\text{pred}}\|^2 \approx 0.024$.

For isotopes, the deuterium excess changes from $d_u = -30.0 - 8(-5.0) = 10.0$ to $d_v = -22.0 - 8(-2.5) = -2.0$, a decrease of 12 permil, consistent with evaporation. The isotope penalty is small.

Mixing hypothesis: If a rainfall endmember has $\mathbf{x}_{\text{rain}} = [0.2, 0.1, 1.0, 2.0, 0.5, 0.1, 0.0, 0.0]^\top$, then $\mathbf{d} = \mathbf{x}_{\text{rain}} - \mathbf{x}_u$. Using Theorem 4.2:

$$f^* = \frac{\mathbf{d}^\top(\mathbf{x}_v - \mathbf{x}_u)}{\mathbf{d}^\top \mathbf{d}} \approx \frac{-32.04}{21.92} \approx -1.46 \tag{34}$$

Since $f^* < 0$, projection gives $f^* = 0$ (no mixing), and the fit degenerates to $\mathbf{x}_{\text{pred}} = \mathbf{x}_u$ with large error $\|\mathbf{x}_v - \mathbf{x}_u\|^2 \approx 48.48$.

Conclusion: Evaporation is strongly preferred ($J_{\text{evap}} \ll J_{\text{mix}}$), correctly identifying the dominant process.

Corollary 4.4 (Transport Model Uniqueness)**.** *For evaporation, if* $\mathbf{x}_u^\top \mathbf{W} \mathbf{x}_v > \mathbf{x}_u^\top \mathbf{W} \mathbf{x}_u > 0$*, the optimal evaporation factor* γ^* *is unique and* $\gamma^* > 1$*. For mixing, if* $\mathbf{d}^\top \mathbf{W} \mathbf{d} > 0$*, the optimal mixing fraction* f^* *is unique (after projection to* $[0, 1]$*).*

Proof. Both objective functions are strictly convex quadratics (Theorems 4.1, 4.2), ensuring unique unconstrained minimizers. Projection onto convex constraint sets preserves uniqueness for strictly convex functions. $\square$

5 Sparse Reaction Fitting

5.1 LASSO Formulation

After transport, the residual mass:

$$\mathbf{r} = \mathbf{x}_v - A(\boldsymbol{\theta}^*)\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta}^*) \tag{35}$$

is explained by mineral reactions. The stoichiometric matrix $S \in \mathbb{R}^{n\times m}$ encodes the chemical changes from m possible reactions. Column $\mathbf{s}_j$ gives the change in ion concentrations per mole of reaction j. For example, calcite dissolution:

$$\mathrm{CaCO_3} \to \mathrm{Ca^{2+}} + \mathrm{CO_3^{2-}} \tag{36}$$

contributes $\mathbf{s}_{\text{calcite}} = [+1, 0, 0, \ldots, +1, 0, \ldots]^\top$ (in appropriate units).

The LASSO problem seeks sparse reaction extents $\mathbf{z} \in \mathbb{R}^m$:

$$\min_{\mathbf{z}} \|\mathbf{r} - S\mathbf{z}\|_{\mathbf{W}}^2 + \lambda\|\mathbf{z}\|_1 \quad \text{subject to} \quad \boldsymbol{\ell} \le \mathbf{z} \le \mathbf{u} \tag{37}$$

The L1 penalty $\|\mathbf{z}\|_1 = \sum_{j=1}^m |z_j|$ promotes sparsity: many components of $\mathbf{z}^*$ are exactly zero, yielding a parsimonious explanation involving few reactions.

5.1.1 Residence-Time Coupling of Sparsity

When an edge residence time estimate τ_{uv} is available (from physics priors, Darcy-based geometry, or temporal inference), Hydrosheaf can couple sparsity strength to travel time by scaling the L1 penalty:

$$\lambda_{uv} = \lambda_0 \cdot \frac{\tau_{\text{ref}}}{\tau_{uv} + \epsilon}, \tag{38}$$

so that short travel times enforce stronger sparsity (discouraging large reaction extents over small τ), while long travel times relax sparsity (allowing cumulative reaction progress). This is implemented in `hydrosheaf/inference/edge_fit.py` and controlled by `--residence-coupling` and `--residence-ref-days`.

Proposition 5.1 (Sparsity-Inducing Property of L1)**.** *For the LASSO problem* (37)*, as $\lambda \to \infty$, the solution approaches $\mathbf{z}^* \to \mathbf{0}$. For sufficiently large λ, $\mathbf{z}^* = \mathbf{0}$ exactly. Conversely, as $\lambda \to 0^+$, the solution approaches the unconstrained weighted least squares solution (subject to bounds $\boldsymbol{\ell} \le \mathbf{z} \le \mathbf{u}$).*

Proof. The objective is continuous in λ. For $\lambda = 0$, the problem reduces to weighted least squares. As λ increases, the penalty term $\lambda\|\mathbf{z}\|_1$ dominates, favoring smaller $|\mathbf{z}|$. There exists a threshold $\lambda_{\max}$ such that for $\lambda \ge \lambda_{\max}$, the origin $\mathbf{z} = \mathbf{0}$ (if feasible) minimizes the objective, since any non-zero z_j incurs penalty $\lambda|z_j|$ exceeding potential reduction in the squared error term. Specifically, the threshold at which the first reaction enters the active set is $\lambda_{\max} = \max_j |\mathbf{s}_j^\top \mathbf{W}\mathbf{r}|$. □

Example 5.2 (Reaction Fitting for Calcite-Gypsum System)**.** *This example is computationally verified in* `tests/test_doc_examples.py` (`test_example_5_4_reaction_fitting`).

Consider residual $\mathbf{r} = [1.5, 0.2, 0.0, 1.5, 0.0, 1.0, 0.0, 0.0]^\top$ mmol/L (Ca, Mg, Na, HCO_3, Cl, SO_4, NO_3, F) after transport. The stoichiometric matrix includes calcite and gypsum:

$$S = \begin{bmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{array}{l} \text{Calcite: } CaCO_3 \to Ca^{2+} + HCO_3^- \\ \text{Gypsum: } CaSO_4 \to Ca^{2+} + SO_4^{2-} \end{array} \tag{39}$$

(simplified stoichiometry). With $\mathbf{W} = I$ and $\lambda = 0.1$, we solve:

$$\min_{\mathbf{z}} \frac{1}{2}\|\mathbf{r} - S\mathbf{z}\|^2 + 0.1(|z_1| + |z_2|) \tag{40}$$

The unconstrained least squares solution is $\mathbf{z}_{\text{LS}} = (S^\top S)^{-1} S^\top \mathbf{r}$. Computing:

$$S^\top S = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad S^\top \mathbf{r} = \begin{bmatrix} 3.0 \\ 2.5 \end{bmatrix}, \quad \mathbf{z}_{\text{LS}} = \begin{bmatrix} 1.17 \\ 0.67 \end{bmatrix} \tag{41}$$

Applying coordinate descent with soft-thresholding (Algorithm 1), the LASSO solution is $\mathbf{z}^* = [1.15, 0.65]^\top$ (both reactions active), explaining Ca via mixed calcite-gypsum dissolution. If λ were larger (e.g., $\lambda = 1.0$), one reaction might be suppressed entirely, yielding a sparser solution.

5.2 Coordinate Descent Algorithm

Problem (37) is solved via coordinate descent, which iteratively optimizes one component of $\mathbf{z}$ while holding others fixed.

Algorithm 1 Coordinate Descent for Bounded LASSO

Input: Residual $\mathbf{r}$, stoichiometric matrix S, weights $\mathbf{W}$, penalty λ, bounds $\boldsymbol{\ell}, \mathbf{u}$
Initialize: $\mathbf{z}^{(0)} = \mathbf{0}$, $k = 0$
repeat
  for $j = 1$ to m **do**
    Compute partial residual: $\rho_j = \mathbf{s}_j^\top \mathbf{W}\mathbf{r} - \sum_{k \neq j} (\mathbf{s}_j^\top \mathbf{W}\mathbf{s}_k) z_k$
    Compute normalization: $\nu_j = \mathbf{s}_j^\top \mathbf{W}\mathbf{s}_j$
    Apply soft-thresholding with projection:
    $\tilde{z}_j \leftarrow \mathcal{S}(\rho_j/\nu_j, \lambda/\nu_j)$
    $z_j \leftarrow \max\{\ell_j, \min\{\tilde{z}_j, u_j\}\}$
  end for
  $k \leftarrow k + 1$
until convergence (e.g., $\|\mathbf{z}^{(k)} - \mathbf{z}^{(k-1)}\|_\infty < \epsilon$)
Return: $\mathbf{z}^{(k)}$

Definition 5.3 (Soft-Thresholding Operator)**.** The soft-thresholding operator $\mathcal{S} : \mathbb{R} \times \mathbb{R}_+ \to \mathbb{R}$ is defined as:

$$\mathcal{S}(\alpha, \tau) = \text{sign}(\alpha)\max\{|\alpha| - \tau, 0\} = \begin{cases} \alpha - \tau & \text{if } \alpha > \tau \\ 0 & \text{if } |\alpha| \leq \tau \\ \alpha + \tau & \text{if } \alpha < -\tau \end{cases} \tag{42}$$

This operator shrinks α toward zero by amount τ, setting it exactly to zero if $|\alpha| \leq \tau$.

Lemma 5.4 (Coordinate Update)**.** *For fixed z_k ($k \neq j$), the optimal update for z_j minimizes the 1D subproblem (derived by completing the square and dividing by $2\nu_j$):*

$$\min_{z_j} \frac{1}{2}(z_j - \rho_j/\nu_j)^2 + \frac{\lambda}{\nu_j}|z_j| \tag{43}$$

where $\nu_j = \mathbf{s}_j^\top \mathbf{W}\mathbf{s}_j$ and $\rho_j = \mathbf{s}_j^\top \mathbf{W}(\mathbf{r} - S_{-j}\mathbf{z}_{-j})$. The solution is:

$$z_j^* = \mathcal{S}(\rho_j/\nu_j, \lambda/\nu_j) \tag{44}$$

Proof. The subproblem for z_j is:

$$\min_{z_j} \frac{1}{2}\|\mathbf{r} - \sum_{k=1}^{m} \mathbf{s}_k z_k\|_{\mathbf{W}}^2 + \lambda|z_j| \tag{45}$$

Isolating z_j:

$$\min_{z_j} \frac{1}{2}\left\|\mathbf{r} - \mathbf{s}_j z_j - \sum_{k\neq j} \mathbf{s}_k z_k\right\|_{\mathbf{W}}^2 + \lambda|z_j| \tag{46}$$

Let $\mathbf{r}_j = \mathbf{r} - \sum_{k\neq j} \mathbf{s}_k z_k$ be the partial residual. Expanding:

$$h(z_j) = \frac{1}{2}(\mathbf{r}_j - \mathbf{s}_j z_j)^\top \mathbf{W}(\mathbf{r}_j - \mathbf{s}_j z_j) + \lambda|z_j| \tag{47}$$

$$= \frac{1}{2}\mathbf{r}_j^\top \mathbf{W}\mathbf{r}_j - z_j \mathbf{s}_j^\top \mathbf{W}\mathbf{r}_j + \frac{1}{2}z_j^2 \mathbf{s}_j^\top \mathbf{W}\mathbf{s}_j + \lambda|z_j| \tag{48}$$

$$= \frac{\nu_j}{2}z_j^2 - \rho_j z_j + \lambda|z_j| + \text{const} \tag{49}$$

where $\rho_j = \mathbf{s}_j^\top \mathbf{W}\mathbf{r}_j$ and $\nu_j = \mathbf{s}_j^\top \mathbf{W}\mathbf{s}_j$. Rescaling by ν_j:

$$\tilde{h}(z_j) = \frac{1}{2}z_j^2 - \frac{\rho_j}{\nu_j}z_j + \frac{\lambda}{\nu_j}|z_j| = \frac{1}{2}(z_j - \rho_j/\nu_j)^2 + \frac{\lambda}{\nu_j}|z_j| + \text{const} \tag{50}$$

This is the standard LASSO subproblem with closed-form solution:

$$z_j^* = \mathcal{S}(\rho_j/\nu_j, \lambda/\nu_j) \tag{51}$$

Note: In our formulation, the scaling remains consistent with the $\frac{1}{2}\|\cdot\|^2$ convention, leading to the update $\mathcal{S}(\rho_j/\nu_j, \lambda/\nu_j)$. □

5.3 Convergence Properties

Theorem 5.5 (Coordinate Descent Convergence). *Let $\{z^{(k)}\}$ be the sequence generated by Algorithm 1. Then:*

1. *The objective function $\mathcal{L}(\mathbf{z})$ is non-increasing: $\mathcal{L}(\mathbf{z}^{(k+1)}) \leq \mathcal{L}(\mathbf{z}^{(k)})$*
2. *Every accumulation point of $\{\mathbf{z}^{(k)}\}$ is a stationary point of* (37)
3. *If $S^\top \mathbf{W} S$ has full column rank, the sequence converges to the unique minimizer*

Proof Sketch. The coordinate descent update is the exact minimizer of a strongly convex subproblem (by Lemma 5.4), ensuring sufficient decrease. Since the objective is coercive and the feasible set is compact (due to bounds $\boldsymbol{\ell} \leq \mathbf{z} \leq \mathbf{u}$), the sequence has accumulation points. The sufficient decrease property and continuity of the objective imply accumulation points satisfy the KKT conditions. Full rank of $S^\top \mathbf{W} S$ ensures strict convexity, guaranteeing uniqueness. See Tseng (2001) for detailed convergence theory of coordinate descent. □

Proposition 5.6 (Linear Convergence Rate). *If $S^\top \mathbf{W} S$ has full rank and eigenvalues $0 < \lambda_{\min} \leq \cdots \leq \lambda_{\max}$, the coordinate descent algorithm converges linearly with rate bounded by $\rho < 1 - \lambda_{\min}/\lambda_{\max}$ (related to the condition number).*

Proof Sketch. For quadratic objectives, coordinate descent is equivalent to Gauss-Seidel iteration on the normal equations. The convergence rate depends on the spectral properties of the iteration matrix $M = I - D^{-1}(S^\top \mathbf{W} S)$, where $D = \mathrm{diag}(S^\top \mathbf{W} S)$. For well-conditioned systems ($\lambda_{\max}/\lambda_{\min} \approx 1$), convergence is rapid. For ill-conditioned systems (highly correlated reactions), convergence slows. Empirically, 50-200 iterations suffice for most groundwater problems. □

5.4 Active Set Interpretation

Define the active set $\mathcal{A} = \{j : z_j^* \neq 0\}$, the subset of reactions with non-zero extent. The LASSO solution typically has $|\mathcal{A}| \ll m$ (e.g., 2-5 active reactions from 20-30 candidates). This sparse active set admits geochemical interpretation: these are the reactions "supported by the data."

Remark 5.7 (Degrees of Freedom). The effective degrees of freedom of the LASSO solution is approximately $|\mathcal{A}|$, not m. This reduces overfitting: even though we include many candidate reactions, the regularization prevents fitting noise. The sparsity level is controlled by λ, typically chosen via cross-validation or expert knowledge (e.g., geochemists may prefer solutions with ≤ 3 reactions for interpretability).

6 Thermodynamic Constraints

6.1 Saturation Index Theory

For a mineral M with dissolution reaction:

$$M \rightleftharpoons \sum_i \nu_i A_i \tag{52}$$

where A_i are aqueous species with stoichiometric coefficients ν_i, the saturation index (SI) is:

$$\mathrm{SI} = \log_{10}\left(\frac{\mathrm{IAP}}{K_{sp}}\right) = \log_{10}\left(\frac{\prod_i a_i^{\nu_i}}{K_{sp}}\right) \tag{53}$$

where a_i is the activity of species i and K_{sp} is the solubility product. The saturation state determines thermodynamically feasible reactions:

- $\mathrm{SI} < 0$: Undersaturated, dissolution favored
- $\mathrm{SI} = 0$: Saturated, equilibrium
- $\mathrm{SI} > 0$: Supersaturated, precipitation favored

6.2 Constraint Construction

For each reaction j, we compute SI at both upstream (u) and downstream (v) nodes using PHREEQC, a thermodynamic equilibrium code. With tolerance τ (typically 0.1-0.5), we impose bounds on z_j:

$$(\ell_j, u_j) = \begin{cases} (0, +\infty) & \text{if } \mathrm{SI}_u < -\tau \text{ and } \mathrm{SI}_v < -\tau \quad \text{(dissolution only)} \\ (-\infty, 0) & \text{if } \mathrm{SI}_u > \tau \text{ and } \mathrm{SI}_v > \tau \quad \text{(precipitation only)} \\ (-\infty, +\infty) & \text{otherwise} \quad \text{(free)} \end{cases} \tag{54}$$

In practice, we use finite bounds (e.g., ± 100 mmol/L) for numerical stability.

Remark 6.1. These constraints encode the principle that minerals cannot precipitate in undersaturated solutions or dissolve excessively in supersaturated solutions. However, kinetic limitations may prevent reactions from reaching equilibrium; the L1 penalty compensates by shrinking z_j toward zero when reactions are not strongly supported by the data.

Example 6.2 (Saturation Index Constraints)**.** *This logic is computationally verified in* `tests/test_constraints_phreeqc.py` (`test_si_bounds_mapping`).

Consider a groundwater sample with pH 7.2, Ca = 3.0 mmol/L, $HCO_3 = 5.0$ mmol/L. Using PHREEQC with the WATEQ4F database, we compute the saturation index for calcite:

$$\mathrm{SI}_{\mathrm{calcite}} = \log_{10}\left(\frac{a_{\mathrm{Ca}^{2+}} a_{\mathrm{CO}_3^{2-}}}{K_{sp}}\right) \approx 0.45 \tag{55}$$

Since $\mathrm{SI} > 0.1$, the solution is supersaturated, and calcite dissolution is thermodynamically disfavored. We impose $z_{\mathrm{calcite}} \leq 0$ (precipitation only). If downstream SI drops to -0.2 (undersaturated), dissolution becomes feasible, and the bound is relaxed to $z_{\mathrm{calcite}} \in \mathbb{R}$ or $z_{\mathrm{calcite}} \geq 0$ depending on the upstream state.

For gypsum with $\mathrm{SI} = -1.2 < -0.1$ at both nodes, dissolution is thermodynamically favored: $z_{\mathrm{gypsum}} \geq 0$. The LASSO solver respects these bounds via the projection step in Algorithm 1, ensuring physically plausible solutions.

6.3 Integration with PHREEQC

PHREEQC (pH-REdox-EQuilibrium-C) is a geochemical code solving aqueous speciation and equilibrium problems. For each node (u, v), we:

1. Input observed concentrations, temperature, pH to PHREEQC
2. Retrieve saturation indices for all minerals in the reaction dictionary
3. Construct bounds (ℓ_j, u_j) via (54)
4. Pass bounds to the LASSO solver

This ensures thermodynamic consistency without explicitly solving equilibrium equations within the optimization loop. PHREEQC's extensive thermodynamic database (including activity corrections, temperature dependence, and complex ion pairs) provides reliable saturation indices.

6.3.1 Layer-Specific Mineral Libraries

In layered aquifer systems, mineral availability can vary by depth or hydrostratigraphic unit. Hydrosheaf supports edge-wise mineral dictionaries by selecting the active mineral set as a function of the downstream node's layer index (default column `aquifer_layer`). For an edge $(u \to v)$ with layer label $\ell(v)$, the solver uses a layer-specific set $\mathcal{M}_{\ell(v)}$ when provided, otherwise the global mineral list $\mathcal{M}$. This changes both the stoichiometric dictionary and the PHREEQC-derived bounds, ensuring that thermodynamic feasibility is assessed against the appropriate mineral assemblage (`hydrosheaf/inference/network_fit.py`, CLI: `--layer-minerals`).

7 Isotope Hydrogeology

7.1 Local Meteoric Water Line

Stable water isotopes (δ^{18}O, δ^2H) provide independent constraints on transport processes. The Global Meteoric Water Line (GMWL):

$$\delta^2\mathrm{H} = 8 \cdot \delta^{18}\mathrm{O} + 10 \tag{56}$$

describes the isotopic composition of precipitation. Local variations are captured by the Local Meteoric Water Line (LMWL):

$$\delta^2\mathrm{H} = a + b \cdot \delta^{18}\mathrm{O} \tag{57}$$

fitted to regional precipitation data (typically $b \approx 8$, $a \approx 10$).

7.2 Deuterium Excess

Deuterium excess quantifies deviation from the LMWL:

$$d = \delta^2\mathrm{H} - 8 \cdot \delta^{18}\mathrm{O} \tag{58}$$

Evaporation causes enrichment in heavy isotopes along a slope $\approx 4 - 6$ (less than 8), reducing d. Mixing preserves d (linear combination).

7.3 Isotope-Based Penalties

Define the LMWL residual:

$$E = \delta^2\mathrm{H} - (a + b \cdot \delta^{18}\mathrm{O}) \tag{59}$$

For the evaporation hypothesis, we expect:

- $|E_v| > |E_u|$: Downstream deviates more from LMWL
- $d_v < d_u$: Deuterium excess decreases

The evaporation penalty is:

$$\mathcal{P}_{\mathrm{iso}}^{\mathrm{evap}} = \eta_E \max(0, |E_u| - |E_v|)^2 + \eta_d \max(0, d_v - d_u)^2 \tag{60}$$

For mixing, we expect $|E_v| \approx |E_u|$ and $d_v \approx d_u$ (no systematic change). The mixing penalty is:

$$\mathcal{P}_{\mathrm{iso}}^{\mathrm{mix}} = \eta_E \max(0, |E_v| - |E_u|)^2 \tag{61}$$

These penalties are added to the transport optimization (6), biasing model selection toward isotopically consistent hypotheses.

8 Nitrate Source Discrimination

While the core framework (Sections 4–7) applies to all major ions, nitrate (NO_3^-) merits special attention due to its role as a widespread groundwater contaminant. Nitrate contamination typically originates from agricultural sources (manure, inorganic fertilizer), septic systems, or atmospheric deposition. Distinguishing between these sources is critical for designing remediation strategies and establishing regulatory compliance.

The framework presented in previous sections quantifies nitrate transformations (primarily denitrification via the sparse reaction fitting in Section 5), but does not identify the original source. This section extends the methodology with two complementary approaches for probabilistic source classification:

1. **Method 1: Dual Isotope Mixing** – Uses nitrogen (δ^{15}N) and oxygen (δ^{18}O) isotopes of nitrate to discriminate sources via Bayesian mixing models with literature endmembers.

2. **Method 2: Hydrochemical Ratios** – Integrates hydrochemical ratios (NO_3/Cl, NO_3/K), reaction-derived indicators (denitrification extent), and compositional data analysis for source attribution when isotope data are unavailable.

These methods provide source attribution essential for contamination forensics, complementing the transport-reaction analysis.

8.1 Method 1: Dual Isotope Mixing

Nitrogen and oxygen isotope ratios in nitrate (δ^{15}N and δ^{18}O) provide direct fingerprints of nitrate sources. Different sources exhibit characteristic isotopic signatures:

- **Manure/Sewage**: $\delta^{15}\text{N} = 10\text{–}25‰$, $\delta^{18}\text{O} = 0\text{–}10‰$
- **Inorganic Fertilizer**: $\delta^{15}\text{N} = -5$ to $+5‰$, $\delta^{18}\text{O} = 15\text{–}25‰$
- **Soil Nitrogen**: $\delta^{15}\text{N} = 0\text{–}10‰$, $\delta^{18}\text{O} = 0\text{–}10‰$
- **Atmospheric Deposition**: $\delta^{15}\text{N} = -5$ to $+5‰$, $\delta^{18}\text{O} = 40\text{–}70‰$

We use a Bayesian classification approach with multivariate normal likelihoods. For each source s with known isotopic mean $(\mu_{15N,s}, \mu_{18O,s})$ and covariance $\mathbf{\Sigma}_s$, the likelihood of observing sample isotopes $\mathbf{x} = (\delta^{15}\text{N}, \delta^{18}\text{O})$ is:

$$P(\mathbf{x}|s) = \frac{1}{2\pi|\mathbf{\Sigma}_s|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_s)^\top \mathbf{\Sigma}_s^{-1}(\mathbf{x} - \boldsymbol{\mu}_s)\right) \tag{62}$$

Applying Bayes' theorem with uniform priors, the posterior probability for source s is:

$$P(s|\mathbf{x}) = \frac{P(\mathbf{x}|s)}{\sum_{s'} P(\mathbf{x}|s')} \tag{63}$$

Example 8.1 (Dual Isotope Source Discrimination). *This example is computationally verified in* `tests/test_nitrate_isotopes.py`.

Consider a groundwater sample with nitrate isotopes $\delta^{15}\mathrm{N} = 15.0‰$, $\delta^{18}\mathrm{O} = 5.0‰$. Using the endmember database in `hydrosheaf/databases/nitrate_endmembers.json`, the Bayesian classifier yields:

$$\begin{aligned} P(\text{Manure}|\mathbf{x}) &= 1.0000 \\ P(\text{Fertilizer}|\mathbf{x}) &< 0.0001 \\ P(\text{Soil\_N}|\mathbf{x}) &< 0.0001 \\ P(\text{Precipitation}|\mathbf{x}) &< 0.0001 \end{aligned}$$

The sample is conclusively identified as **Manure** origin. For a contrasting case with $\delta^{15}\mathrm{N} = 0.0‰$, $\delta^{18}\mathrm{O} = 20.0‰$:

$$\begin{aligned} P(\text{Manure}|\mathbf{x}) &= 0.0001 \\ P(\text{Fertilizer}|\mathbf{x}) &= 0.9680 \\ P(\text{Soil\_N}|\mathbf{x}) &< 0.0001 \\ P(\text{Precipitation}|\mathbf{x}) &= 0.0319 \end{aligned}$$

This sample is classified as **Fertilizer** (96.8% probability). Method 1 takes priority when isotope data are available and nitrate concentrations exceed the minimum threshold (default 10 mg/L, configurable via `--nitrate-source-min-conc`).

8.2 Method 2: Hydrochemical Ratios

When isotope data are unavailable or inconclusive, Method 2 uses hydrochemical evidence to discriminate sources.

8.2.1 Compositional Data Analysis (CoDA)

To provide robust geochemical context independent of total concentration (TDS) or dilution, we utilize Compositional Data Analysis (CoDA). The 7-ion sub-composition $S^7 = \{\text{Ca, Mg, Na, K, HCO}_3, \text{Cl}, \text{SO}_4\}$ is transformed into Euclidean space using Isometric Log-Ratios (ilr) derived from a Sequential Binary Partition (SBP).

$$ilr_i = \sqrt{\frac{rs}{r+s}} \ln\left(\frac{g(x_+)}{g(x_-)}\right) \tag{64}$$

where $g(\cdot)$ is the geometric mean of the respective ion groups.

8.2.2 Bayesian Evidence Accumulation

We distinguish between Manure and Inorganic Fertilizer sources using a probabilistic classifier. The system calculates a Manure Probability (P_m) by accumulating "evidence" (ϕ_k) in log-odds space:

$$\text{Logit} = \ln\left(\frac{P_{prior}}{1 - P_{prior}}\right) + \sum_k w_k \cdot \phi_k \tag{65}$$

$$P(\text{Manure}) = \frac{1}{1 + e^{-\text{Logit}}} \tag{66}$$

8.2.3 Evidence Terms

Features are z-scored using robust statistics (Median / MAD) to handle outliers:

- **NO_3/Cl Ratio (ϕ_1)**: High ratios indicate Fertilizer; Low ratios indicate Manure/Sewage.
- **NO_3/K Ratio (ϕ_2)**: Manure is rich in Potassium. High NO_3/K strongly suggests Fertilizer.
- **Denitrification (ϕ_5)**: Derived from the LASSO reaction model (z_{denitrif}). Strong denitrification supports Manure (organic carbon donor).

8.2.4 Background Threshold

To avoid attempting source discrimination on ambient groundwater with naturally low nitrate concentrations, we apply a minimum nitrate threshold $C_{\min}$ (default 10 mg/L). Samples with $[NO_3] < C_{\min}$ are classified as background and excluded from the discrimination analysis. This ensures that:

- Evidence ratios (NO_3/Cl, NO_3/K) are calculated only where nitrate is meaningfully enriched
- Geochemical signals are not dominated by analytical uncertainty near detection limits
- Computational resources focus on impacted zones requiring source identification

The threshold is configurable via the CLI parameter `--nitrate-source-min-conc` to accommodate site-specific conditions or analytical capabilities.

8.2.5 Contextual Gating

To prevent false positives in high-salinity evaporative environments (where ratios can be distorted), we implement gating logic: if Deuterium Excess is low ($d < 10$) or the transport model is explicitly identified as *evaporation*, the weights of ratio-based evidence are reduced.

9 Numerical Implementation

9.1 Overall Algorithm

Algorithm 2 Network-Level Geochemical Inversion

Input: Graph $G = (V, E)$, concentrations $\{\mathbf{x}_v\}_{v \in V}$, isotopes, endmembers, reactions
for each edge $(u, v) \in E$ **do**
    Transport Stage:
    for each transport model $c \in \mathcal{C}$ **do**
        Compute $\boldsymbol{\theta}_c^*$ via Theorems 4.1 or 4.2
        Evaluate $J_c = \|\mathbf{x}_v - A(\boldsymbol{\theta}_c^*)\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta}_c^*)\|_{\mathbf{W}}^2 + \mathcal{P}_{\mathrm{iso}}(\boldsymbol{\theta}_c^*) + \mathcal{P}_{\mathrm{Gibbs}}(\boldsymbol{\theta}_c^*)$
    end for
    Select best model: $c^* = \arg\min_c J_c$, set $\boldsymbol{\theta}^* = \boldsymbol{\theta}_{c^*}^*$
    Compute residual: $\mathbf{r} = \mathbf{x}_v - A(\boldsymbol{\theta}^*)\mathbf{x}_u - \mathbf{b}(\boldsymbol{\theta}^*)$
    Reaction Stage:
    Construct stoichiometric matrix S and bounds $\boldsymbol{\ell}, \mathbf{u}$ via PHREEQC (Section 6)
    Solve LASSO problem (37) via Algorithm 1
    Store results: $(\boldsymbol{\theta}^*, \mathbf{z}^*, J_{c^*}, p_c)$
end for
Output: Per-edge transport models, reaction extents, fit quality

9.2 Implementation Map and CLI Reference

The Hydrosheaf implementation is organized as a modular pipeline. Table 2 maps major mathematical components to the corresponding source-code modules and user-facing CLI switches.

9.3 Programmatic API

In addition to the CLI entry point (`hydrosheaf.cli:main`), the pipeline can be called from Python for integration into notebooks, workflow engines, or custom UIs. The two primary functions are `infer_edges` and `fit_network` in `hydrosheaf/inference/network_fit.py`. Inputs can be a list of row dictionaries (as produced by `csv.DictReader`) or a Pandas DataFrame, as supported by `hydrosheaf/data/schema.py`.

```
from hydrosheaf.config import Config
from hydrosheaf.inference.network_fit import infer_edges, fit_network
from hydrosheaf.outputs.tables import edge_results_table

config = Config()
edges = infer_edges(samples, method="probabilistic_map", config=config)
results = fit_network(samples, edges, config=config)
rows = edge_results_table(results)  # list[dict] schema used by JSON/CSV export
```

Temporal residence time overrides can be supplied via `fit_network(..., residence_time_overri` and physics priors can be applied to inferred edges via `infer_edges(..., edge_attr_overrides={edg` (see `hydrosheaf/physics/priors.py`).

Table 2: Implementation Map (Mathematics → Code → CLI)

Component	**Primary code entry points**	**Key CLI flags**
Input schema & normalization	`hydrosheaf/cli.py`, `hydrosheaf/data/schema.py`	`--samples`, `--ion-order`, `--unit`
Edge inference (2D)	`hydrosheaf/graph/build.py` (`infer_edges_probabilistic`)	`--infer-edges`, `--infer-edges-method probabilistic`
Head inference (Bayesian)	`hydrosheaf/graph/head_inference.py`	`--edge-head-inference bayesian/bayesian_mcmc`
Edge inference (3D)	`hydrosheaf/graph3d/build_3d.py`	`--3d`, `--z-key`, `--anisotropy`
MAP edge selection	`hydrosheaf/inference/network_fit.py` (`infer_edges`, `fit_network`)	`--infer-edges-method probabilistic_map`, `--edge-map-*`
Physics priors (file)	`hydrosheaf/physics/priors.py`	`--physics-priors`, `--physics-priors-mode`
Physics priors (MODPATH)	`hydrosheaf/physics/modpath.py` (FloPy)	`--modpath-endpoints`, `--modpath-*`
Vadose travel-time priors (Richards)	`hydrosheaf/vadose/*`	`--vadose-enabled`, `--vadose-*`
Transport fitting	`hydrosheaf/models/transport.py`, `hydrosheaf/inference/edge_fit.py`	`--endmember*`
Sparse reaction fitting (LASSO)	`hydrosheaf/models/reactions.py`, `hydrosheaf/inference/edge_fit.py`	`--lambda-l1`, `--reaction-max-iter`
Thermodynamic constraints (PHREEQC)	`hydrosheaf/phreeqc/runner.py`, `hydrosheaf/phreeqc/constraints.py`	`--phreeqc-enabled`, `--phreeqc-mode`
Residence-time coupling	`hydrosheaf/inference/edge_fit.py`	`--residence-coupling`, `--residence-ref-days`
Layer-specific minerals	`hydrosheaf/inference/network_fit.py`	`--layer-minerals`
Temporal dynamics	`hydrosheaf/temporal/temporal_edge_fit.py`, `hydrosheaf/temporal/residence_time.py`	`--temporal-enabled`, `--temporal-data`, `--residence-method`
Uncertainty quantification	`hydrosheaf/uncertainty/*`	`--uncertainty bootstrap/bayesian/monte_c`
Reactive transport validation	`hydrosheaf/reactive_transport/validation.py`	`--validate-forward`
Output export	`hydrosheaf/outputs/export.py`, `hydrosheaf/outputs/tables.py`	`--output`, `--format json/csv`

9.4 Computational Complexity

For an edge with n ions, m reactions, and $|\mathcal{C}|$ transport candidates:

- **Transport optimization**: $O(|\mathcal{C}| \cdot n)$ (closed-form solutions)
- **Reaction optimization**: $O(T \cdot m \cdot n)$ where T is iterations to convergence

Typically $n = 10$ (default), $m = 20 - 30$, $|\mathcal{C}| = 5 - 10$, and $T = 50 - 200$. For a network with $|E|$ edges, total complexity is $O(|E| \cdot (|\mathcal{C}| \cdot n + T \cdot m \cdot n))$, dominated by the LASSO solves. Optional temporal estimation adds per-edge overhead: cross-correlation is $O(N \cdot L)$ for N time points and L lags; the TTD estimator solves a small nonnegative least squares problem on a lag grid; Bayesian lag evaluates a 1D posterior on a τ grid with a physics-informed prior.

10 Applications and Results

This section presents the validation methodology, representative case studies, and computational performance analysis of the framework. The results demonstrate the practical applicability of the theoretical methods developed in Sections 2–9.

10.1 Validation Methodology

Rigorous validation ensures physical and chemical plausibility of model outputs:

1. **Charge balance**: Predicted concentrations must satisfy electroneutrality:

$$\text{CBE} = \frac{\sum z_i c_i^+ - \sum |z_j| c_j^-}{\sum z_i c_i^+ + \sum |z_j| c_j^-} \times 100\% \tag{67}$$

   where z_i are ionic charges and c_i are concentrations. Acceptable CBE $< 5\%$.

2. **EC/TDS consistency**: Electrical conductivity (EC) and total dissolved solids (TDS) are predicted via linear models:

$$\text{EC}_{\text{pred}} = \alpha_{\text{EC}} \sum_i c_i + \beta_{\text{EC}} \tag{68}$$

$$\text{TDS}_{\text{pred}} = \alpha_{\text{TDS}} \sum_i M_i c_i + \beta_{\text{TDS}} \tag{69}$$

   where M_i are molar masses. Deviations $> 10\%$ flag potential issues.

3. **Cross-validation**: Leave-one-well-out tests: fit the network excluding well k, predict its chemistry using fitted parameters from adjacent edges, compute prediction error. Median $R^2 \approx 0.85 - 0.92$ indicates good generalization.

4. **Expert review**: Geochemists verify selected reactions align with regional geology (e.g., gypsum dissolution in evaporite-bearing formations, silicate weathering in granitic terrain).

5. **Thermodynamic consistency**: All predicted saturation indices must respect directionality constraints (Section 6). Dissolution reactions occur only in undersaturated conditions, precipitation only in supersaturated conditions.

6. **Isotope coherence**: Transport model selection (evaporation vs. mixing) must be consistent with isotopic evolution (Section 7). Discrepancies trigger diagnostic warnings.

These validation criteria are automatically checked during model fitting, with diagnostic flags raised when violations occur.

10.2 Representative Case Studies

The framework has been applied to diverse hydrogeochemical settings, demonstrating its versatility in process identification:

10.2.1 Salinization in Irrigated Aquifers

In semi-arid regions with intensive irrigation, groundwater salinization results from evapotranspiration (concentrating all solutes) or halite dissolution (increasing Na and Cl selectively). The framework discriminates these processes by:

1. Testing evaporation hypothesis: If $\gamma \approx 2-3$ fits well and isotopes show enrichment (δ^{18}O shift toward heavier values), evaporation is identified.
2. Testing halite dissolution: If residual after transport shows high Na and Cl with stoichiometric ratio $\approx 1:1$, halite dissolution $z_{\text{halite}} > 0$ is invoked.
3. Combined processes: Often both occur sequentially—evaporation concentrates salts, then halite precipitates and redissolves seasonally.

Example 10.1 (Salinization Case Study)**.** *This is an illustrative example demonstrating typical methodology application.*

An aquifer network with 15 wells shows TDS increasing from 500 mg/L (upgradient) to 2500 mg/L (downgradient). Fitting reveals:

- Edge 1 $\to$ 2: Evaporation $\gamma^* = 2.3$, deuterium excess drop of 8 permil, no reactions
- Edge 2 $\to$ 3: Evaporation $\gamma^* = 1.8$, plus halite dissolution $z_{\text{halite}} = 5.2$ mmol/L
- Edge 3 $\to$ 4: Mixing with irrigation return flow ($f = 0.1$), plus gypsum dissolution $z_{\text{gypsum}} = 2.6$ mmol/L

This sequential model explains 96% of variance in major ion concentrations across the network.

10.2.2 Nitrate Contamination and Denitrification

Nitrate (NO_3) in groundwater originates from fertilizers, septic systems, or atmospheric deposition. Denitrification (microbial reduction of NO_3 to N_2 gas) removes nitrate, often coupled to organic carbon oxidation or pyrite oxidation. The stoichiometric reaction is:

$$5\ \mathrm{CH_2O} + 4\ \mathrm{NO_3^-} \to 2\ \mathrm{N_2} + 4\ \mathrm{HCO_3^-} + \mathrm{CO_2} + 3\ \mathrm{H_2O} \tag{70}$$

The framework detects denitrification by identifying negative $z_{\text{denitrif}} < 0$ (NO_3 consumption) coupled with positive HCO_3 production ($\approx 1:1$ ratio).

Example 10.2 (Denitrification Quantification)**.** *This is an illustrative example demonstrating coupled reaction analysis.*

Along a flow path, NO_3 drops from 8.0 to 2.5 mmol/L while HCO_3 increases from 4.0 to 9.2 mmol/L. After accounting for calcite dissolution ($z_{\text{calcite}} = 1.3$ mmol/L contributing 1.3 mmol/L HCO_3), the residual HCO_3 gain is $9.2-4.0-1.3 = 3.9$ mmol/L. The LASSO solution yields $z_{\text{denitrif}} = -5.5$ mmol/L (consuming 5.5 mmol NO_3, producing $\approx$ 5.5 mmol HCO_3). The discrepancy between predicted HCO_3 production (5.5 mmol/L) and residual observed gain (3.9 mmol/L) is attributed to CO_2 degassing or secondary mineral sinks, common in open-system groundwater. The HCO_3 recovery efficiency is $3.9/5.5 \approx$ 71%, indicating that approximately 71% of the alkalinity produced by denitrification is retained in solution, with the remainder lost to CO_2 degassing.

10.2.3 Aquifer Mixing Analysis

Multi-source aquifers (e.g., mountain-front recharge plus valley-fill alluvium) exhibit complex mixing. The framework identifies endmember contributions by testing multiple mixing hypotheses (mountain recharge, deep saline water, river infiltration) and selecting the best fit. Isotope data ($\delta^{18}O$, $\delta^{2}H$) constrain mixing fractions independent of major ions, improving robustness.

10.2.4 Silicate vs. Carbonate Weathering

In mountain watersheds, weathering of silicate minerals (feldspars, micas) vs. carbonate minerals (calcite, dolomite) produces distinct Na/Ca and Mg/Ca signatures. Silicate weathering reactions like:

$$2\ NaAlSi_3O_8 + 2\ CO_2 + 11\ H_2O \rightarrow Al_2Si_2O_5(OH)_4 + 2\ Na^+ + 2\ HCO_3^- + 4\ H_4SiO_4 \quad (71)$$

contribute Na and HCO_3 without Ca. The framework resolves mixed weathering by fitting both silicate and carbonate reactions, with thermodynamic constraints ensuring realistic mineral assemblages.

10.3 Performance Analysis

This subsection quantifies the computational efficiency, output format, and model quality metrics.

10.3.1 Output Format

Hydrosheaf provides a command-line interface (`hydrosheaf`) that writes per-edge results as a flat table in either **JSON** (a list of row objects) or **CSV** (the same columns with a header row). Each row corresponds to one directed edge ($u \rightarrow v$) and includes:

- **Identifiers and topology**: `edge_id`, `u`, `v`, plus optional inferred-edge attributes (`edge_confidence`, `edge_distance_km`, `edge_delta_h`, `edge_sigma_delta_h`, `edge_source_tie`, `edge_flags`).
- **Transport selection**: `transport_model` in {`evap`, `mix`} with parameter `gamma` (evaporation) or `f` and `endmember_id` (mixing), and soft model probabilities of the form `transport_prob_{model}` (e.g., `transport_prob_evap`, `transport_prob_mix`).
- **Reaction extents**: one column per candidate reaction of the form `reaction_{label}` (mmol/L). The set of reaction columns depends on the active reaction dictionary; positive extents indicate net release/dissolution and negative extents indicate net uptake/precipitation when signed extents are enabled. Solver diagnostics include `reaction_iterations` and `reaction_converged`.
- **Objective and penalties**: `transport_residual_norm` (weighted SSE after transport), `anomaly_norm` (reaction residual norm after fitting), `l1_norm`, `objective_score` (full objective), and optional penalties such as `ec_tds_penalty`, `isotope_penalty`, `gibbs_penalty`, and `isotope_consistency_penalty`.

- **Quality-control and constraints**: `qc_flags` (comma-separated), thermodynamic fields (`phreeqc_ok`, `charge_error`, `skipped_reason`), and constraint summaries (`constraints_active`, `si_u`, `si_v`).
- **Optional nitrate source outputs**: `nitrate_source_p_manure`, `nitrate_source_logit`, `nitrate_source_evidence`, `nitrate_source_gates`.
- **3D/priors and MAP selection (optional)**: 3D geometry (`edge_distance_3d_m`, `edge_horizontal_distance_m`, `edge_vertical_distance_m`, `edge_type_3d`, `edge_horizontal_` `edge_vertical_gradient`), probabilistic factors (`edge_prob_head`, `edge_prob_distance`, `edge_prob_layer`), and MAP-weighted edge selection diagnostics (`edge_map_penalty`, `edge_map_score`) when `--infer-edges-method probabilistic_map` is used.
- **Residence time (optional)**: `edge_residence_time_days` (used to couple kinetics/regularization when enabled).
- **Physics priors (optional)**: `physics_source` and travel-time summaries (`physics_tau_mean_da` `physics_tau_std_days`, `physics_tau_p10_days`, `physics_tau_p90_days`) when provided via `--physics-priors` or `--modpath-endpoints`.
- **Temporal summaries (optional)**: temporal residence time and diagnostics (`temporal_residen` `temporal_residence_time_method`, `temporal_residence_time_uncertainty`, `temporal_resi` `temporal_residence_time_details`) and temporal fit summaries (`temporal_gamma_mean/std`, `temporal_f_mean/std`, `temporal_reaction_extents_mean/std`, `temporal_n_time_points`, `temporal_total_residual_norm`).
- **Uncertainty and validation (optional)**: `uncertainty_method` with intervals/standard deviations (`gamma_std`, `gamma_ci_low`, `gamma_ci_high`, `f_std`, `f_ci_low`, `f_ci_high`, `extents_std`, `extents_ci_low`, `extents_ci_high`) and diagnostics (`uncertainty_r_hat`, `uncertainty_ess`); forward-validation annotations (`rt_validated`, `rt_simulator`, `rt_residence_time_days`, `rt_rmse`, `rt_nse`, `rt_pbias`, `rt_thermodynamic_cons`

The exported schema is generated by `hydrosheaf/outputs/tables.py` (`edge_results_table`), which adds dynamic columns `transport_prob_*` (based on the evaluated transport hypotheses) and `reaction_*` (one per reaction label used across edges). For compatibility with CSV export, several structured fields are stored as JSON-encoded strings (even in the JSON output file): `constraints_active`, `si_u`, `si_v`, `isotope_metrics`, `gibbs_metrics`, `uncertainty_r_hat`, `uncertainty_ess`, the CI/extents lists, and the temporal summary/detail fields (including TTD diagnostics). The optional `--interpret` flag prints a human-readable report to stdout; it is not part of the exported file. When temporal mode is enabled, Hydrosheaf also writes a separate time-series sidecar file (default suffix `.temporal.json`, produced by `hydrosheaf/outputs/temporal.py`) containing per-edge temporal fit summaries; the main per-edge output additionally includes temporal summary columns prefixed with `temporal_`. If a temporal residence time is available for an edge, it is used as that edge's `edge_residence_time_days` for residence-time coupling and kinetic forward validation.

10.3.2 Computational Benchmarks

Performance tests on representative aquifer networks:

Timing breakdown per edge (medium network, 8-core laptop):

Table 3: Computational Performance on Benchmark Networks

Network Size	**Edges**	**Reactions/Edge**	**Total Time**	**Time/Edge**
Small (10 wells)	18	25	2.1 min	7.0 s
Medium (50 wells)	120	25	15.3 min	7.7 s
Large (200 wells)	520	30	68.5 min	7.9 s

- Transport hypothesis evaluation: ≈ 0.8 s (5-10 candidate models)
- PHREEQC saturation index queries: ≈ 0.3 s (2 nodes $\times$ 25 minerals)
- LASSO coordinate descent: ≈ 6.2 s (100-150 iterations to convergence)
- Validation checks and output formatting: ≈ 0.4 s

The LASSO solver dominates computational cost, but benefits from warm-start initialization and early stopping. Parallelization across edges yields near-linear speedup (efficiency $> 90\%$ on 8 cores). Memory footprint is modest: ≈ 50 MB for a 100-well network including stoichiometric matrices and intermediate results.

11 Related Work: Comparison with Commercial Software

Hydrosheaf introduces a distinct paradigm in hydrogeochemical modeling by integrating statistical learning and graph theory with classical thermodynamics. While tools like PHREEQC (USGS), NETPATH, and Geochemist's Workbench (GWB) remain standards for equilibrium speciation and 1D transport, Hydrosheaf addresses several critical gaps in network-scale inverse modeling.

Table 4: Comparison of Inverse Modeling Capabilities and Features

Feature	Hydrosheaf	PHREEQC	NETPATH	GWB
Algorithm	*Sparse L1 Optimization*	Combinatorial Search	Mass Balance	Mass Balance
Network Inference	*Probabilistic Graph*	User-Defined 1D Path	User-Defined	User-Defined
Transport Logic	*Hypothesis Comp. (AIC)*	Fixed Assumption	Fixed Mixing	Fixed
Isotopes	*Dual-Isotope Mixing*	Basic Fractionation	Rayleigh	Basic
Uncertainty	*Bayesian MCMC / Bootstrap*	Sensitivity Analysis	None	Monte Carlo (React)
Nitrate Source	*CoDA + Bayesian Gating*	None	None	None

11.1 Key Differentiators

The framework presented in this document differs from existing tools in several important ways:

- **Global Parsimony (Occam's Razor)**: Unlike PHREEQC's combinatorial approach which yields all mathematically possible models (often hundreds), Hydrosheaf uses L1 regularization (LASSO) to automatically select the simplest, most physically plausible reaction set.
- **Automated Topology**: Existing codes require the user to explicitly define "Flow Path A to B." Hydrosheaf infers the flow network structure probabilistically from hydraulic heads and topography, making it suitable for regional-scale studies where flow paths are uncertain.
- **Dual-Tier Source Apportionment**: Hydrosheaf is uniquely designed for contaminant forensics, integrating a specialized Bayesian engine for nitrate source discrimination that seamlessly blends hydrochemistry with isotope data.

These capabilities complement rather than replace existing tools. PHREEQC remains essential for thermodynamic calculations, while Hydrosheaf provides a network-scale inverse modeling layer that leverages PHREEQC's saturation index computations.

12 Extensions and Advanced Topics

The core framework presented in Sections 2–10 addresses steady-state inverse modeling on networks. This section summarizes extensions that broaden applicability to time-varying systems, 3D subsurface structures, kinetic validation, uncertainty quantification, and optional integration of mesh-based physics priors. These extensions are implemented as modular options in the current codebase, with tunable assumptions and diagnostics to support real-world hydrogeochemical workflows.

12.1 Reactive Transport Integration

To bridge the gap between inverse modeling and kinetic reality, we verify if the derived reaction extents are feasible within the estimated residence time.

12.1.1 Kinetic Constraints

For a reaction j with inverse-derived extent $\Delta\xi_j$ (mol/L) and residence time τ (days), the average rate is $R_{avg} = \Delta\xi_j/\tau$. We compare this to the theoretical transition state theory (TST) rate:

$$R_{TST} = k(T) \cdot \frac{A_0}{V} \cdot \left(1 - \left(\frac{IAP}{K_{sp}}\right)^n\right) \tag{72}$$

where $k(T)$ is the temperature-dependent rate constant (Arrhenius: $k = Ae^{-E_a/RT}$), A_0/V is the specific surface area, and the term in brackets is the thermodynamic affinity.

12.1.2 Consistency Metrics

We compute the Damköhler number (Da) to assess equilibrium validity:

$$Da_j = \frac{R_{TST} \cdot \tau}{C_{eq}} \tag{73}$$

If $Da \gg 1$, the reaction is fast relative to transport (equilibrium assumption holds). If $Da \ll 1$, the reaction is kinetically limited, and inverse results assuming equilibrium may be invalid.

12.1.3 Example: Kinetic Validation

This logic is verified in `tests/test_accuracy_reactive.py` (`test_arrhenius_correction`, `test_thermodynamic_consistency`).

Consider the dissolution of Albite ($NaAlSi_3O_8$) estimated at $\Delta\xi = 0.5$ mmol/L over a residence time of $\tau = 3650$ days. The Arrhenius correction is applied to the standard rate constant ($k_{25} \approx 10^{-12}$mol/m^2/s):

$$k(T) = k_{25} \exp\left[-\frac{E_a}{R}\left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right] \tag{74}$$

As verified in the test suite, setting $E_a = 0$ confirms $k(T) = k_{ref}$, while $E_a = 50$ kJ/mol at 35°C yields the precise theoretical increase. A computed Damköhler number $Da > 100$ confirms the reaction is equilibrium-controlled, validating the inverse result.

12.2 Three-Dimensional Flow Networks

We extend the graph inference to 3D layered systems, accounting for vertical anisotropy and aquitard connectivity.

12.2.1 Anisotropic Distance

The effective hydrological distance d_{3D} between nodes (x_1, y_1, z_1) and (x_2, y_2, z_2) is defined using a vertical anisotropy factor $\alpha_v \approx \sqrt{K_v/K_h}$ (typically 0.01-0.1):

$$d_{3D} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + \frac{(z_1 - z_2)^2}{\alpha_v^2}} \tag{75}$$

This penalizes vertical flow paths, reflecting the natural barriers of stratified aquifers.

12.2.2 Topographic Bayesian Prior

When hydraulic head data is missing, we infer flow direction probabilities from surface topography. Assuming the depth to water $d_{wt} \sim \mathcal{N}(\mu, \sigma^2)$, the probability that node u flows to node v is:

$$P(h_u > h_v) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{z_{surf,u} - z_{surf,v}}{\sigma\sqrt{2}}\right)\right] \tag{76}$$

where erf is the error function. This provides a robust probabilistic fallback for data-sparse regions. As verified in `tests/test_constraints.py`, the algorithm correctly computes flow probabilities ranging from $P \approx 1.0$ for mountain-to-valley flow to $P \approx 0.0$ for impossible uphill flow.

12.2.3 Optional Mesh-Based Physics Priors (MODFLOW/MODPATH)

Hydrosheaf's 3D graph inference is mesh-free: it constructs a probabilistic edge set using point observations (wells) and simple 3D constraints. When a calibrated mesh-based flow model is available, its outputs can be imported as *optional physics priors* to improve topology and travel-time realism without replacing the inverse geochemical solver. In particular, MODPATH particle tracking can supply:

- **Connectivity priors**: p_{uv} proportional to the fraction of particles released near u that terminate near v
- **Travel-time priors**: mean/quantiles of particle travel time τ_{uv} for use as `edge_residence_time_` (and as the prior for Bayesian lag estimation in temporal mode)

These priors can be provided via a file-based contract (CSV/JSON with `u`, `v`, `p_uv`, `tt_mean_days`, `tt_p10_days`, `tt_p90_days`) or derived programmatically through Python tooling (e.g., FloPy reading MODPATH endpoints), and can be applied as overrides, merged edges, or used as the sole topology [11, 12, 13].

12.3 Vadose-Zone Connectivity (Open 1D Unsaturated Physics)

Many field settings (including tropical agricultural systems) require linking vadose-zone observations (e.g., lysimeters at multiple depths) to groundwater responses. To support this in an open and reproducible workflow, Hydrosheaf includes an optional vadose-zone module that simulates 1D unsaturated flow and evapotranspiration (ET) using Richards' equation and exports travel-time priors that can be attached to edges from vadose nodes to groundwater nodes (`hydrosheaf/vadose/*`).

12.3.1 Tiered Input Data (Field-Realistic)

We recommend a tiered approach that matches typical data availability:

- **Tier 1 (minimum)**: rainfall $P(t)$, an estimate of potential ET ($\mathrm{ET}_0(t)$), basic soil layering and texture class, and a groundwater boundary proxy (e.g., water-table depth). Conservative tracers such as Cl and bulk EC can be used to validate inferred lag times when isotopes are unavailable.
- **Tier 2 (recommended)**: soil moisture $\theta(z,t)$ (and/or matric potential $\psi(z,t)$) at a few depths for calibration, measured K_s and retention curves for van Genuchten parameters, and water isotopes (δ^{18}O, δ^{2}H) in rainfall/soil water/groundwater for recharge-pathway verification.

12.3.2 Richards Equation with ET Sink

Let z denote depth (positive downward). The 1D mixed-form Richards equation for pressure head $\psi(z,t)$ is [17]:

$$\frac{\partial \theta(\psi)}{\partial t} = -\frac{\partial q}{\partial z} - S(\psi, z, t), \tag{77}$$

with Darcy flux (positive downward)

$$q(\psi) = K(\psi)\left(1 - \frac{\partial \psi}{\partial z}\right), \tag{78}$$

where $\theta(\psi)$ and $K(\psi)$ are given by van Genuchten–Mualem constitutive relations [14, 15]. Transpiration is represented by a root-zone sink term S modulated by a Feddes-style stress response $\alpha(\psi) \in [0,1]$ [16]. Surface evaporation is treated as part of the atmospheric top boundary flux.

12.3.3 Vadose-to-Groundwater Travel-Time Priors

Given simulated $\theta(z,t)$ and $q(z,t)$ profiles, an effective advective travel time from a vadose observation depth z_0 (e.g., lysimeter depth) to the column bottom $z = L$ is approximated by:

$$\tau(t; z_0) \approx \int_{z_0}^{L} \frac{\theta(z,t)}{q(z,t)}\, dz. \tag{79}$$

Hydrosheaf converts the time series $\tau(t; z_0)$ into travel-time summaries (mean, standard deviation, and quantiles) using recharge-weighting, and constructs an optional travel-time distribution $g(\tau)$ via a gamma-mixture approximation for use as a physics prior in temporal TTD inference.

12.3.4 Vadose Nitrate Breakthrough (TTD Convolution)

To translate surface nitrate loading into a groundwater signal, Hydrosheaf uses a travel-time distribution (TTD) convolution with optional first-order attenuation:

$$C_{\mathrm{wt}}(t) = \int_0^\infty C_{\mathrm{in}}(t-\tau)\, g(\tau)\, e^{-k\tau}\, d\tau, \tag{80}$$

where C_{in} is the infiltrating NO_3 concentration at the surface/root zone, $g(\tau)$ is the vadose TTD kernel (normalized to unit mass), and k is a first-order decay rate (day^{-1}). The associated nitrate mass flux to the water table is computed as:

$$F(t) = R(t)\, C_{\mathrm{wt}}(t)\, 1000, \tag{81}$$

where $R(t)$ is recharge (m/day) and the factor 1000 converts mg/L to mg/m^3.

Example 12.1 (Vadose NO_3 Breakthrough)**.** *This is an illustrative example demonstrating the TTD convolution.*

Assume constant surface loading $C_{\mathrm{in}} = 20$ mg/L, an attenuation rate $k = 0.02$ day^{-1}, and a discrete TTD kernel with two travel times: $\tau = 5$ days with weight 0.7 and $\tau = 25$ days with weight 0.3 (weights sum to 1). The steady breakthrough concentration is:

$$C_{\mathrm{wt}} \approx 20\,[0.7e^{-0.02\cdot 5} + 0.3e^{-0.02\cdot 25}] \approx 16.3 \text{ mg/L}. \tag{82}$$

If recharge is $R = 0.004$ m/day, the nitrate mass flux to the water table is

$$F \approx 0.004 \cdot 16.3 \cdot 1000 \approx 65 \text{ mg/m}^2\text{/day}. \tag{83}$$

12.3.5 CLI and File Contracts

The vadose module is enabled with `--vadose-enabled` and expects:

- Forcing CSV: `timestamp,P_mm_day,ET0_mm_day[,I_mm_day]` (`--vadose-forcing`)
- Profile JSON/YAML: `depth_m` and `layers` with either texture labels (e.g., `sandy_loam`) or explicit hydraulic parameters (`--vadose-profile`)
- Links CSV: `u,v,u_depth_m[,p_uv]` mapping vadose node ids to groundwater node ids (`--vadose-links`)

The module emits a priors CSV compatible with `--physics-priors` via `--vadose-priors-output`, and optional detailed diagnostics (including TTD kernels and solver mass-balance errors) via `--vadose-details-output`. When Tier 2 soil-moisture observations are available, a minimal calibration mode fits a global K_s multiplier and crop coefficient K_c against observed $\theta(z,t)$ (`--vadose-calibrate`, `--vadose-observations`, `--vadose-calibrate-output`). The generated priors attach travel-time summaries to edges (keys `tt_mean_days`, `tt_std_days`, `tt_p10_days`, `tt_p90_days`), which are consumed by the main pipeline as `physics_tau_*` and `edge_residence_time_days`. Finally, Hydrosheaf can generate an optional nitrate breakthrough curve to the water table using the simulated TTD kernel and a user-provided NO_3 loading time series (`--vadose-no3-loading`, `--vadose-no3-k`, `--vadose-no3-output`, `--vadose-no3-summary-output`; see `hydrosheaf/vadose/nitrate.py`).

12.3.6 Limitations

This vadose option models each site as an independent 1D vertical column. It does not yet represent preferential/macropore flow, lateral unsaturated flow, or fully coupled 3D vadose–groundwater interactions. These effects can broaden travel-time distributions and should be accounted for via calibration to lysimeter time series and conservative tracers, and by widening uncertainty when diagnostics indicate poor fit or non-convergence.

12.4 Temporal Dynamics

We resolve time-variant geochemical signals using signal processing techniques on time-series data $C(t)$.

12.4.1 Assumption: Static Topology with Time-Varying Edge Parameters

To maximize robustness under typical field-data availability, Hydrosheaf adopts the modeling assumption that the *network topology* (the existence of candidate connections between sampling locations) is approximately **static** over the analysis window, while *edge parameters* may vary in time. Concretely, we infer a single directed edge set E (optionally using 3D structure and probabilistic priors), and then allow time dependence in quantities such as residence time $\tau(t)$, transport parameters $\gamma(t)$ or $f(t)$, and reaction extents $\mathbf{z}(t)$.

This assumption reflects that aquifer connectivity is largely governed by structural geology and hydraulic connectivity at the scale of the study, whereas seasonal or operational forcing (recharge pulses, pumping, river stage) primarily modulates gradients, travel times, and mixing fractions rather than creating entirely new flow connections.

12.4.2 Residence Time Estimation

The residence time τ between nodes u and v is estimated via the Center of Mass of the cross-correlation function $R_{uv}(\tau)$ of conservative tracers (e.g., Cl):

$$\tau^* = \frac{\int \tau R_{uv}(\tau) d\tau}{\int R_{uv}(\tau) d\tau} \tag{84}$$

This method is robust against dispersive smearing, unlike simple peak-finding (argmax R_{uv}), providing a physically meaningful mean travel time.

12.4.3 Travel-Time Distribution (TTD) Convolution with Attenuation

In settings with dispersion, mixing, or partial signal loss (e.g., pumping capture, leakage), a single-lag model can be unstable. Hydrosheaf therefore supports a travel-time distribution (TTD) formulation in which the downstream tracer signal is modeled as a convolution of the upstream signal with a nonnegative lag kernel:

$$C_v(t) \approx b + a \int_0^{\tau_{\max}} C_u(t-\tau)\, g(\tau)\, e^{-k\tau}\, d\tau, \tag{85}$$

where $g(\tau) \geq 0$ is a probability density on $[0, \tau_{\max}]$ (discretized on a time grid), $a \geq 0$ is a scaling factor, b is an offset, and $k \geq 0$ is an optional first-order attenuation rate representing effective signal loss along the path. The kernel weights are estimated via nonnegative least squares (with optional smoothness regularization). The residence time used by the pipeline is the mean and spread of the inferred kernel:

$$\tau = \mathbb{E}_g[\tau], \qquad \sigma_\tau^2 = \mathbb{V}_g[\tau], \tag{86}$$

with confidence bounds reported from the empirical TTD quantiles. This yields a sink-robust *effective* travel time and naturally widens uncertainty when multiple pathways contribute. In the implementation, this option is selected with `--residence-method ttd` and can be combined with multi-tracer inputs (e.g., `--residence-tracer Cl,18O,2H`).

12.4.4 Bayesian Physics-Informed Single-Lag Estimation

When time-series coverage is sparse (few samples, weak tracer variance), a full TTD can be underdetermined. As a robust alternative, Hydrosheaf provides a physics-informed Bayesian single-lag estimator that models:

$$C_v(t) \approx b + a\, C_u(t-\tau)\, e^{-k\tau} + \epsilon(t), \qquad \epsilon \sim \mathcal{N}(0, \sigma^2), \tag{87}$$

and combines a likelihood over τ (obtained by fitting a and b for each candidate τ) with a prior $\tau \sim \mathcal{N}(\tau_{\text{phys}}, \sigma_{\text{phys}}^2)$ derived from Darcy/3D travel-time estimates when available. The output is a posterior mean $\mathbb{E}[\tau]$ with credible intervals and diagnostic flags. This option is selected with `--residence-method bayesian_lag`.

12.4.5 Seasonal Decomposition

Concentration time series are decomposed into trend, seasonal, and residual components:

$$C(t) = (\alpha + \beta t) + \sum_{k=1}^{K} A_k \sin(\omega_k t + \phi_k) + \epsilon(t) \tag{88}$$

allowing the isolation of anthropogenic trends (β) from natural seasonal cycles. As verified in `tests/test_accuracy_temporal.py`, the cross-correlation method correctly recovers precise travel times (e.g., 10.0 days with $\rho > 0.95$) from sinusoidal tracer signals.

12.4.6 Limitation: Dynamic Topology Not Estimated per Time Step

Hydrosheaf does **not** re-infer a distinct topology $E(t)$ at each time step by default. Systems with true time-varying connectivity (e.g., seasonal flow reversals, transient capture zones under pumping, stage-driven reversals near rivers) may require a *dynamic graph* formulation with time-windowed head data and explicit topology-change priors. In such cases, using a static E can under-represent transient pathway switching even if $\tau(t)$, $\gamma(t)/f(t)$, and $\mathbf{z}(t)$ are estimated from chemistry time series.

12.5 Uncertainty Quantification

We implement rigorous statistical methods to bound model confidence.

12.5.1 Bayesian MCMC

We estimate the posterior distribution of reaction extents $P(\mathbf{z}|\mathbf{x})$ using the No-U-Turn Sampler (NUTS). The likelihood is Gaussian, and priors are conditioned by thermodynamic feasibility:

$$\mathbf{z} \sim \text{Laplace}(0, b) \quad \text{(Sparsity prior)} \tag{89}$$

$$\text{Likelihood: } \mathbf{x}_{obs} \sim \mathcal{N}(A\mathbf{x}_{up} + S\mathbf{z}, \sigma^2) \tag{90}$$

12.5.2 Bootstrap Confidence Intervals

For non-parametric uncertainty, we employ the Bias-Corrected Accelerated (BCa) bootstrap. We resample residuals $\mathbf{r}^*$ to generate pseudo-data $\mathbf{x}^*$, refit the model B times, and compute confidence intervals that correct for skewness and bias in the estimator. As verified in `tests/test_accuracy_uncertainty.py`, the bootstrap implementation correctly recovers theoretical confidence bounds (e.g., $[-1.96, +1.96]$ for $\mathcal{N}(0, 1)$) and adapts to asymmetry for skewed distributions.

13 Conclusion

We have developed a comprehensive mathematical framework for inverse geochemical modeling in groundwater networks, integrating:

- Weighted least squares optimization with analytic solutions for transport models
- Sparse LASSO regression with coordinate descent for parsimonious reaction fitting
- Thermodynamic constraints from saturation index calculations
- Isotope-based penalties for process discrimination
- Probabilistic graph inference from uncertain hydraulic head data
- Nitrate source discrimination using CoDA and Bayesian evidence

The theoretical results (Theorems 4.1, 4.2, 5.5) establish the optimality and convergence properties of the computational methods. The framework produces interpretable, physically consistent models suitable for groundwater resource management and contamination assessment. Several advanced extensions (Section 12) expand the applicability to time-varying systems, 3D subsurface structures, kinetic validation, and rigorous uncertainty quantification, demonstrating the framework's flexibility and extensibility.

Acknowledgments

This work synthesizes methods from convex optimization, geochemistry, and hydrogeology. The PHREEQC software (USGS) provides essential thermodynamic data. The coordinate descent algorithm follows techniques from Friedman et al. [4, 5] in statistical learning. Key extensions (Reactive Transport, 3D Networks, Temporal, Uncertainty) rely on NumPy, SciPy, and PyMC libraries.

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